

Level 1 — Commands You Will Use Every Day

These are the commands you should master first.

Command	Purpose	Example	Output
pwd	Show current directory	pwd	/home/zico
ls	List files and folders	ls	project db.sqlite3
ls -l	Detailed listing	ls -l	permissions, owner, size
ls -a	Show hidden files	ls -a	.bashrc .gitignore
cd	Change directory	cd project	Changes location
cd ..	Go up one directory	cd ..	Moves back
cd ~	Go to home directory	cd ~	/home/user
mkdir	Create folder	mkdir myproject	Creates folder
rmdir	Remove empty folder	rmdir olddir	Folder removed
touch	Create file	touch test.txt	File created
cp	Copy file	cp a.txt b.txt	File copied
cp -r	Copy folder	cp -r app backup	Folder copied
mv	Move or rename	mv file.txt new.txt	Renamed
rm	Delete file	rm file.txt	Deleted
rm -r	Delete folder	rm -r oldproject	Folder removed
clear	Clear screen	clear	Blank terminal
history	Show command history	history	Previous commands

Level 2 — Reading Files

Useful when viewing logs and configuration files.

Command	Purpose	Example
cat	Display file content	cat settings.py
less	Read long files	less access.log
more	Similar to less	more file.txt
head	First lines	head -20 file.txt
tail	Last lines	tail file.txt
tail -f	Live monitoring	tail -f nginx.log
wc	Count lines/words	wc file.txt
nl	Number lines	nl file.txt

Level 3 — Finding Things

Extremely useful on servers.

Command	Purpose	Example
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find	Search files	find . -name "*.py"
locate	Fast search	locate nginx.conf
which	Find executable	which python3
whereis	Find binary/docs	whereis nginx
grep	Search text	grep DEBUG settings.py
grep -r	Recursive search	grep -r SECRET_KEY .

Examples:

```
grep "ERROR" server.log
find /home -name "*.log"
```

Level 4 — File Editing

Essential for VPS management.

Command	Purpose
nano	Beginner editor
vim	Advanced editor
vi	Older vim version

Example:

```
nano settings.py
```

Save:

```
Ctrl + O
Enter
```

Exit:

```
Ctrl + X
```

Level 5 — Permissions

One of the most important VPS topics.

Command	Purpose
chmod	Change permissions
chown	Change owner
chgrp	Change group
umask	Default permissions

Examples:

```
chmod 755 manage.py
```

Level 6 — User Management

For server administration.

Command	Purpose
whoami	Current user
id	User information
adduser	Create user
passwd	Change password
su	Switch user
sudo	Execute as administrator
groups	Show groups
last	Login history

Examples:

```
whoami
sudo nano /etc/nginx/nginx.conf
```

Level 7 — Process Management

Critical for Django deployment.

Command	Purpose
ps	List processes
ps aux	Detailed process list
top	Real-time monitoring
htop	Better top
kill	Stop process
kill -9	Force stop
jobs	Background jobs
bg	Resume in background
fg	Bring to foreground

Examples:

```
ps aux
kill 1234
```

Level 8 — Package Management (Ubuntu)

You will use these constantly.

Command	Purpose
apt update	Update package list
apt upgrade	Upgrade packages
apt install	Install package
apt remove	Remove package
apt autoremove	Clean unused packages
dpkg -l	List installed packages

Examples:

```
sudo apt update
sudo apt install nginx
```

Level 9 — Networking

Very important for VPS administration.

Command	Purpose
ip a	Show IP addresses
hostname	Show hostname
ping	Test connectivity
curl	Request URL
wget	Download files
netstat	Network connections
ss	Modern netstat
dig	DNS lookup
nslookup	DNS lookup

Examples:

```
ping google.com
curl https://example.com
```

Level 10 — Disk Management

Useful when server storage becomes full.

Command	Purpose
df -h	Disk usage
du -sh	Folder size
lsblk	Disk list
mount	Mounted disks
fdisk -l	Disk partitions

Examples:

```
df -h
du -sh project/
```

Level 11 — Archives and Backups

Command	Purpose
tar	Archive files
gzip	Compress
gunzip	Extract gzip
zip	Create zip
unzip	Extract zip

Examples:

```
tar -czvf backup.tar.gz project/
tar -xzvf backup.tar.gz
```

Level 12 — SSH (The VPS Commands)

These are the commands that actually connect you to servers.

Command	Purpose
ssh	Connect to server
scp	Copy files
rsync	Synchronize files
ssh-keygen	Create SSH key
ssh-copy-id	Copy public key

Examples:

```
ssh root@123.45.67.89
scp project.zip root@server:/home/root/
```

Level 13 — Services (Django Deployment)

You'll use these almost every day on a VPS.

Command	Purpose
systemctl start	Start service
systemctl stop	Stop service
systemctl restart	Restart service
systemctl status	Service status
systemctl enable	Auto-start
journalctl	View service logs

Examples:

```
sudo systemctl restart nginx
sudo systemctl status nginx
sudo journalctl -u nginx
```

Level 14 — Django Deployment Commands

Commands you'll repeatedly use during deployment.

Command	Purpose
<code>python3 -m venv venv</code>	Create virtual environment
<code>source venv/bin/activate</code>	Activate environment
<code>pip install</code>	Install packages
<code>pip freeze</code>	Generate requirements
<code>python manage.py migrate</code>	Apply migrations
<code>python manage.py collectstatic</code>	Collect static files
<code>gunicorn</code>	Run production server

Example:

```
source venv/bin/activate
pip install gunicorn
```

Level 15 — Log Analysis (Advanced VPS Skill)

When your website breaks, these commands save you.

Command	Purpose
<code>tail -f</code>	Follow logs live
<code>journalctl -xe</code>	System errors
<code>grep ERROR</code>	Find errors
<code>dmesg</code>	Kernel messages

Examples:

```
tail -f /var/log/nginx/error.log
journalctl -xe
```

The 20 Commands I Would Memorize First

If your objective is Django deployment on Ubuntu, memorize these first:

```
pwd
ls
cd
mkdir
touch
cp
mv
rm
cat
nano
find
grep
whoami
sudo
ps
top
apt
ip a
ssh
systemctl
```

If you become comfortable with just these 20 commands, you'll already be capable of navigating a Linux server, installing software, editing configuration files, connecting to a VPS, troubleshooting services, and deploying a Django application. The remaining commands will become easier because they build on the same concepts.